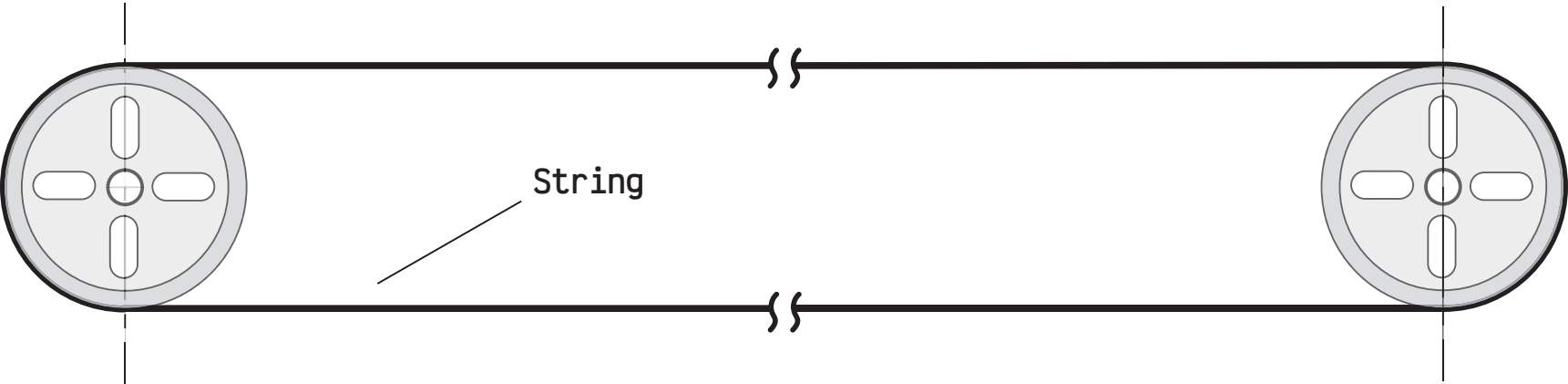String-Actuated System: approximate modeling